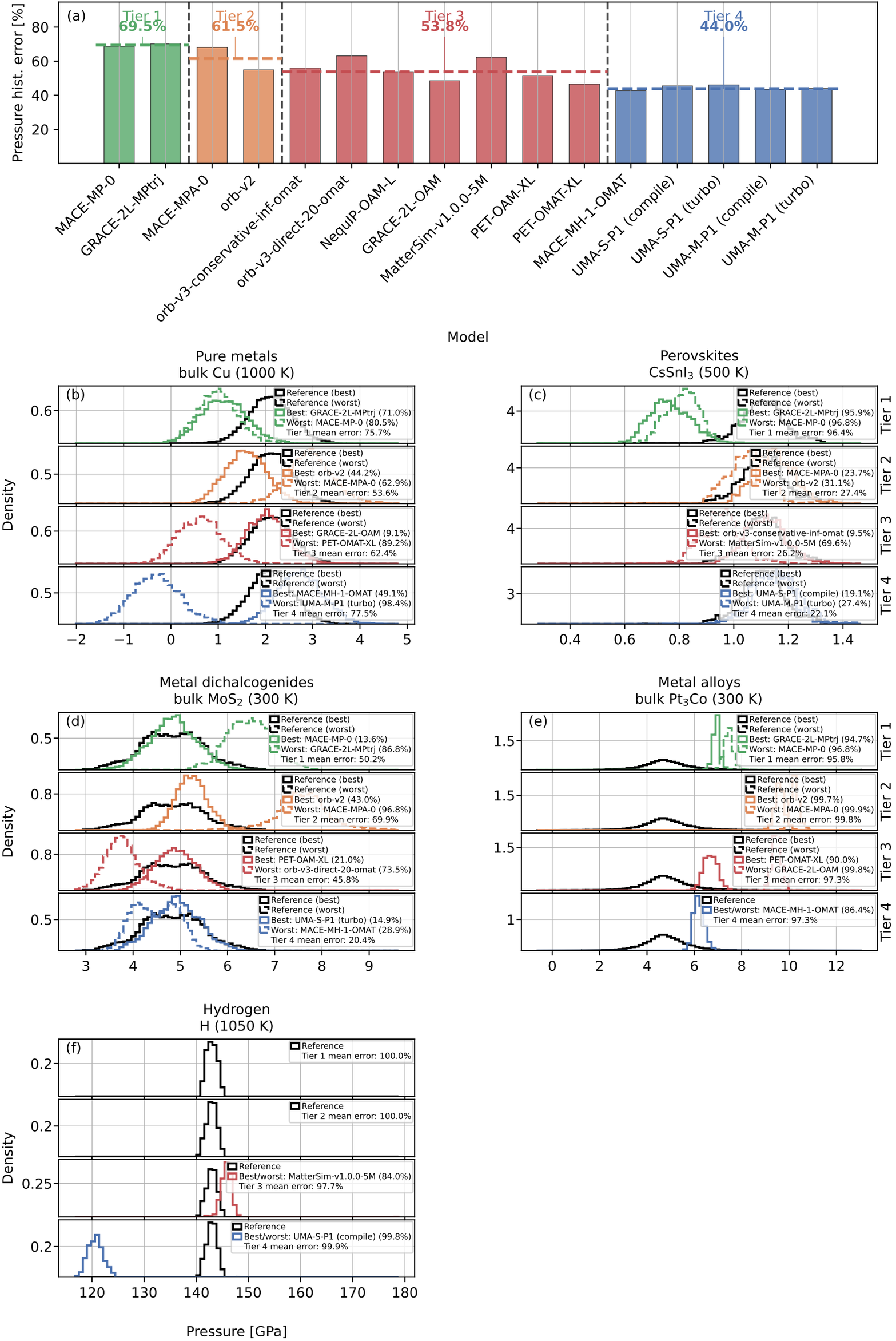

(a)
Tier 1
69.5%
Tier 2
61.5%
Tier 3
53.8%
Tier 4
44.0%
Pressure hist. error [%]
MACE-MP-0
GRACE-2L-MPtrj
MACE-MPA-0
orb-v2
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
NequIP-OAM-L
GRACE-2L-OAM
MatterSim-v1.0.0-5M
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1 (compile)
UMA-S-P1 (turbo)
UMA-M-P1 (compile)
UMA-M-P1 (turbo)
Model
Pure metals
bulk Cu (1000 K)
(b)
Reference (best)
Reference (worst)
Best: GRACE-2L-MPtrj (71.0%)
Worst: MACE-MP-0 (80.5%)
Tier 1 mean error: 75.7%
Best: orb-v2 (44.2%)
Worst: MACE-MPA-0 (62.9%)
Tier 2 mean error: 53.6%
Best: GRACE-2L-OAM (9.1%)
Worst: PET-OMAT-XL (89.2%)
Tier 3 mean error: 62.4%
Best: MACE-MH-1-OMAT (49.1%)
Worst: UMA-M-P1 (turbo) (98.4%)
Tier 4 mean error: 77.5%
Density
Perovskites
CsSnI3 (500 K)
(c)
Best: GRACE-2L-MPtrj (95.9%)
Worst: MACE-MP-0 (96.8%)
Tier 1 mean error: 96.4%
Best: MACE-MPA-0 (23.7%)
Worst: orb-v2 (31.1%)
Tier 2 mean error: 27.4%
Best: orb-v3-conservative-inf-omat (9.5%)
Worst: MatterSim-v1.0.0-5M (69.6%)
Tier 3 mean error: 26.2%
Best: UMA-S-P1 (compile) (19.1%)
Worst: UMA-M-P1 (turbo) (27.4%)
Tier 4 mean error: 22.1%
Metal dichalcogenides
bulk MoS2 (300 K)
(d)
Best: MACE-MP-0 (13.6%)
Worst: GRACE-2L-MPtrj (86.8%)
Tier 1 mean error: 50.2%
Best: orb-v2 (43.0%)
Worst: MACE-MPA-0 (96.8%)
Tier 2 mean error: 69.9%
Best: PET-OAM-XL (21.0%)
Worst: orb-v3-direct-20-omat (73.5%)
Tier 3 mean error: 45.8%
Best: UMA-S-P1 (turbo) (14.9%)
Worst: MACE-MH-1-OMAT (28.9%)
Tier 4 mean error: 20.4%
Metal alloys
bulk Pt3Co (300 K)
(e)
Best: GRACE-2L-MPtrj (94.7%)
Worst: MACE-MP-0 (96.8%)
Tier 1 mean error: 95.8%
Best: orb-v2 (99.7%)
Worst: MACE-MPA-0 (99.9%)
Tier 2 mean error: 99.8%
Best: PET-OMAT-XL (90.0%)
Worst: GRACE-2L-OAM (99.8%)
Tier 3 mean error: 97.3%
Best/worst: MACE-MH-1-OMAT (86.4%)
Tier 4 mean error: 97.3%
Hydrogen
H (1050 K)
(f)
Reference
Tier 1 mean error: 100.0%
Tier 2 mean error: 100.0%
Best/worst: MatterSim-v1.0.0-5M (84.0%)
Tier 3 mean error: 97.7%
Best/worst: UMA-S-P1 (compile) (99.8%)
Tier 4 mean error: 99.9%
Pressure [GPa]